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(57) Abstract :
This invention introduces a large-scale, real-time forecasting architecture based on Kernel Adaptive Filters (KAF) to predict stock mid-prices across all NIFTY 50 constituents. The system leverages advanced algorithms such as Kernel Least Mean Squares (KLMS), Kernel Recursive Least Squares (KRLS), and Quantized Kernel Least Mean Squares (QKLMS), each selected for their ability to adapt dynamically to evolving market conditions. By integrating these algorithms into a multi-stock parallel processing pipeline, the architecture ensures scalability and efficiency, enabling simultaneous forecasting across diverse securities. A key innovation lies in its robustness under non-stationary market microstructure conditions, where traditional linear models often fail. The adaptive kernel framework continuously updates its parameters in response to shifting statistical properties, capturing nonlinear dependencies and high-frequency variations inherent in financial data. This allows for more accurate mid-price predictions, which are critical for applications such as algorithmic trading, risk management, and liquidity provision. The architecture is designed for real-time deployment, ensuring low latency and high throughput, essential for modern trading environments. Its modular design also allows for seamless integration with existing financial systems, making it a practical solution for institutions seeking enhanced predictive capabilities. Overall, this invention represents a significant advancement in adaptive financial forecasting technology.

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